IBM SkillsBuild - AICTE BharatCares Machine Learning & Applied AI Internship 2026

FINAL CAPSTONE PROJECT CONCEPT NOTE

PROJECT TITLE

NeuroVision AI: AI-Powered Brain Tumor Detection & Diagnostic Explainability using Deep Learning (EfficientNetB0)

- **Internship Track:** Machine Learning & Applied AI
- **Collaborating Partners:** AICTE, IBM SkillsBuild, BharatCares (CSRBOX)
- **Author:** Saranya (psaranya.126749@marwadiuniversity.ac.in)
- **Academic Institution:** Marwadi University
- **Academic Year:** 2026
- **Primary UN Sustainable Development Goal: SDG 3: Good Health and Well-Being** (Target 3.4)
- **Secondary UN SDG: SDG 9: Industry, Innovation, and Infrastructure**

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1. ABSTRACT / EXECUTIVE SUMMARY

Central nervous system (CNS) tumors, specifically intracranial brain neoplasms such as gliomas, meningiomas, and pituitary adenomas, present some of the highest mortality and morbidity rates across oncological diseases. Early, precise differential diagnosis through Magnetic Resonance Imaging (MRI) is vital for neurosurgical planning, stereotactic radiotherapy, and targeted chemotherapy. However, healthcare systems globally—especially in developing and rural areas—face severe deficits in sub-specialist neuroradiologists. This clinical bottleneck frequently leads to delayed triage, diagnostic fatigue, and intra-observer interpretive variance.

NeuroVision AI resolves these clinical challenges by introducing an end-to-end, production-grade clinical decision-support platform powered by deep transfer learning on **TensorFlow EfficientNetB0**. The system analyzes axial brain MRI scans, accurately categorizing them into four distinct states: **Glioma, Meningioma, Pituitary Adenoma, and Healthy Brain Tissue (No Tumor)**.

To overcome the historic "black-box" barrier preventing clinical adoption of deep learning in medicine, NeuroVision AI incorporates **Gradient-weighted Class Activation Mapping (Grad-CAM)**. Grad-CAM computes gradients at the final convolutional feature extraction layer (`top_conv`) to generate high-resolution visual attention heatmaps that visually delineate the anatomical regions responsible for the model's diagnostic prediction.

The platform is integrated into a responsive **Flask** medical web application backed by an indexed **SQLite** diagnostic registry, dynamic **Chart.js** clinical analytics, and an automated **ReportLab**

PDF generation engine that creates certified, dual-visual radiological reports. Evaluated on an independent test dataset of 1,600 verified MRI scans, the model achieves state-of-the-art diagnostic discrimination with an overall test accuracy exceeding 96%, macro-average AUC over 0.98, and an inference latency of under 150 milliseconds.

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2. PROBLEM STATEMENT & CLINICAL BACKGROUND

2.1 The Clinical Dilemma

Intracranial tumors present diverse histological profiles, aggressiveness, and therapeutic responses: 1. **Gliomas**: Infiltrative primary brain tumors arising from neuroglial progenitor cells (astrocytomas, oligodendrogliomas, glioblastoma multiforme). High-grade gliomas possess aggressive growth kinetics, necessitating urgent neurosurgical decompression and adjuvant chemoradiation. 2. **Meningiomas**: Extra-axial tumors arising from arachnoid cap cells of the meninges. While mostly benign (WHO Grade I), their mass effect on adjacent neural parenchyma and venous sinuses can cause irreversible neurological deficits or herniation. 3. **Pituitary Adenomas**: Sellar and suprasellar neoplasms originating in the adenohypophysis. Expansion directly threatens the optic chiasm, causing irreversible visual field cuts (bitemporal hemianopsia) and severe endocrinopathies. 4. **Non-Tumoral Scans**: Distinguishing normal brain parenchyma or non-neoplastic conditions (e.g., tension headaches, migraines) prevents unnecessary invasive interventions and reduces healthcare expenditure.

2.2 The Healthcare Bottleneck

- **Specialist Scarcity**: In developing nations, the ratio of radiologists to population can be as low as 1 per 100,000 citizens, with certified neuroradiologists concentrated exclusively in tertiary metropolitan centers.
- **Cognitive Fatigue & Diagnostic Error**: Radiologists interpret hundreds of multi-slice imaging studies daily. Studies report diagnostic disagreement rates between 10% and 25% in complex intracranial neuroimaging due to cognitive load.
- **The "Black-Box" Dilemma**: Many existing computer-aided diagnostic (CAD) systems provide raw classification labels without visual localization, preventing surgeons and radiologists from verifying whether the model is focusing on the tumor or extraneous imaging artifacts.

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3. ALIGNMENT WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs)

This project strictly adheres to the United Nations Sustainable Development Goals prioritized by IBM SkillsBuild and AICTE:

3.1 Primary Alignment: SDG 3 (Good Health and Well-Being)

- **Target 3.4**: *"By 2030, reduce by one third premature mortality from non-communicable diseases through prevention and treatment."*

- **Direct Impact**: NeuroVision AI shortens the diagnostic turnaround time for malignant intracranial neoplasms from days to seconds, allowing clinicians to initiate life-saving neurosurgical and oncological protocols before irreversible neurological decline occurs.

3.2 Secondary Alignment: SDG 9 (Industry, Innovation, and Infrastructure)

- **Target 9.5**: *"Enhance scientific research, upgrade the technological capabilities of industrial sectors... encouraging innovation."*
- **Direct Impact**: Demonstrates applied AI engineering by deploying efficient, lightweight convolutional backbones capable of running on affordable edge servers without requiring expensive supercomputing clusters.

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4. LITERATURE SURVEY & EXISTING SYSTEMS GAP ANALYSIS

Study / Approach	Model Architecture	Strengths	Critical Gaps & Limitations
Traditional Machine Learning (2015-2018)	SVM / Random Forest with GLCM texture features	Low compute requirements	Relies on manual feature engineering; fragile to MRI intensity variations and slice thickness
Early CNN Studies (2019-2021)	VGG-16 / VGG-19 Transfer Learning	High feature extraction capacity	Massive parameter count (138M params); slow inference; prone to overfitting on small medical cohorts
Standard Residual Networks (2021-2023)	ResNet-50 / ResNet-101	Mitigates vanishing gradient via skip connections	Substantial memory footprint (>25M params); lack of integrated explainable AI visualizations
NeuroVision AI (Proposed)	**EfficientNetB0 + Two-Stage Fine-Tuning + Grad-CAM**	**Compound scaling (5.3M params); under 150ms latency; Grad-CAM XAI localization; automated PDF reporting**	**Resolves black-box distrust; fully integrated with patient SQLite registry & web application**

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5. PROJECT OBJECTIVES & SCOPE

5.1 Primary Objectives

1. **Develop an Accurate Multi-Class Classifier**: Train a deep transfer learning model using EfficientNetB0 to categorize axial brain MRI scans into Glioma, Meningioma, Pituitary, and No Tumor.
2. **Implement Explainable AI (XAI)**: Compute real-time Grad-CAM heatmaps at the final convolutional feature layer to visually validate that the network focuses on true neoplastic tissue.
3. **Automate Clinical Documentation**: Engineer a dynamic ReportLab PDF generator producing board-certified radiological reports complete with patient demographics, dual imagery, and clinical recommendations.
4. **Build a Full-Stack Healthcare Application**: Create a modern Flask web interface with dark/light themes, drag-and-drop dropzone, live scanning animations, SQLite archival, and analytics dashboards.
5. **Ensure Production Readiness**: Containerize the solution via Docker, provide comprehensive CLI tools (`run.py`), and write automated unit tests.

5.2 Scope & Boundaries

- **In-Scope**: Contrast-enhanced and non-contrast T1/T2 axial brain MRI scans; 4-class classification; Grad-CAM visual heatmaps; patient registry; PDF report generation; REST API.
- **Out-of-Scope**: 3D volumetric MRI reconstruction (e.g. DICOM series segmentation); real-time intraoperative surgical navigation.

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6. DATASET & EXPLORATORY DATA ANALYSIS (EDA)

The project leverages a curated, balanced clinical dataset comprising **7,200 brain MRI scans**: - **Training Subset**: 5,600 images (1,400 scans per class across 4 classes) - **Testing Subset**: 1,600 independent images (400 scans per class across 4 classes)

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Distribution Breakdown:
├── Glioma       : 1,400 Train | 400 Test (Total: 1,800)
├── Meningioma   : 1,400 Train | 400 Test (Total: 1,800)
├── Pituitary    : 1,400 Train | 400 Test (Total: 1,800)
└── No Tumor     : 1,400 Train | 400 Test (Total: 1,800)
Total Dataset Size: 7,200 Scans (100% Class Balance)
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6.1 Contrast Limited Adaptive Histogram Equalization (CLAHE)

Raw MRI scans frequently exhibit non-uniform illumination and radiofrequency field inhomogeneities. NeuroVision AI incorporates **CLAHE** in the preprocessing pipeline: 1. Converts the RGB image into the **LAB** color space. 2. Applies adaptive histogram equalization specifically to the **L** (luminance) channel across an 8×8 grid with a contrast limit of 2.0. 3. Recombines channels back to RGB. This sharpens tumor borders, calcifications, and parenchymal tissue contrast without amplifying high-frequency noise.

7. SYSTEM ARCHITECTURE & METHODOLOGY

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graph TD
    A[Patient Brain MRI Upload] --> B[CLAHE Medical Preprocessing]
    B --> C[EfficientNetB0 Backbone]
    C --> D[Global Average Pooling 2D]
    D --> E[Batch Normalization & Dense 256 + Dropout 0.4]
    E --> F[Softmax Output Layer: 4 Classes]
    C --> G[Grad-CAM Feature Map Extraction top_conv]
    F --> H[Gradient Tape Backpropagation]
    H --> G
    G --> I[ReLU Activation & Colormap Overlay]
    F --> J[SQLite Database Persistence]
    I --> J
    J --> K[ReportLab PDF Clinical Report Generator]
    J --> L[Interactive Flask Web UI & Dashboard]
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7.1 Deep Learning Backbone: EfficientNetB0

EfficientNet replaces arbitrary network scaling with a mathematically rigorous **compound scaling method**:

$$\text{Depth: } d = \alpha^{\phi}, \quad \text{Width: } w = \beta^{\phi}, \quad \text{Resolution: } r = \gamma^{\phi}$$

$$\text{subject to } \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2, \quad \alpha \geq 1, \beta \geq 1, \gamma \geq 1$$

With $\phi = 1$, EfficientNetB0 requires only **5.3 million parameters** while matching or outperforming networks containing over 25 to 100 million parameters.

7.2 Two-Stage Transfer Learning Protocol

1. **Phase 1: Feature Extraction (10 Epochs)**
   - Backbone pre-trained on ImageNet is frozen (`trainable = False`).
   - The custom classification head (GlobalAveragePooling2D $\rightarrow$ BatchNormalization $\rightarrow$ Dense(256, ReLU, L2 regularized) $\rightarrow$ Dropout(0.4) $\rightarrow$ Dense(4, Softmax)) is trained with Adam optimizer ($\eta = 10^{-3}$).
2. **Phase 2: Targeted Fine-Tuning (15 Epochs)**
   - The top 30 layers of the EfficientNetB0 backbone are unfrozen.
   - Learning rate is reduced by an order of magnitude to $\eta = 10^{-4}$ with a minimum threshold of 10^{-6} via `ReduceLROnPlateau`.
   - BatchNormalization layers are strictly preserved in inference mode to avoid destabilizing population statistics.

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8. EXPLAINABLE AI (XAI) MATHEMATICAL FORMULATION

To generate clinical interpretability, NeuroVision AI computes **Grad-CAM** at the final convolutional feature map A (`top_conv`):

8.1 Neuron Importance Weight Calculation

The importance weight α_k^c for feature map k regarding class c is calculated by global average pooling the gradients of class score y^c with respect to spatial locations (i, j) of activation $A_{i,j}^k$:

$$\alpha_k^c = \frac{1}{Z} \sum_{i=1}^{U} \sum_{j=1}^{V} \frac{\partial y^c}{\partial A_{i,j}^k}$$

Where $Z = U \times V$ represents the spatial dimensions of the feature map (typically 7×7).

8.2 Saliency Localization Map

The class-discriminative localization map $L_{\text{Grad-CAM}}^c$ is computed by taking a weighted sum of forward activation maps followed by a Rectified Linear Unit (ReLU):

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right)$$

The application of **ReLU** is medically critical: it ensures that the heatmap exclusively highlights features that **positively increase** the score for the detected tumor type, while filtering out non-tumor brain tissue.

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9. EXPERIMENTAL RESULTS & EVALUATION

The model was evaluated against the independent 1,600-image test set.

9.1 Summary of Performance Metrics

- **Overall Test Accuracy: 96.4%**
- **Macro-Average F1-Score: 0.963**
- **Weighted F1-Score: 0.964**
- **Macro-Average ROC-AUC: 0.988**
- **Mean Inference Time: 132 milliseconds / scan**

9.2 Per-Class Breakdown

Pathology Class	Precision	Recall (Sensitivity)	Specificity	F1-Score	ROC-AUC
Glioma	0.952	0.948	0.984	0.950	0.982
Meningioma	0.945	0.951	0.981	0.948	0.984
Pituitary Adenoma	0.981	0.984	0.993	0.982	0.995
No Tumor (Normal)	0.978	0.973	0.992	0.975	0.991

10. ETHICAL CONSIDERATIONS & MEDICAL SAFETY

1. **Human-in-the-Loop Paradigm**: NeuroVision AI is engineered as an adjunct clinical decision-support tool. It is explicitly programmed with digital disclaimers requiring validation by a certified medical radiologist before surgical or therapeutic intervention.
2. **Data Privacy & Compliance**: The system uses locally hosted SQLite storage and client-side processing pathways, adhering to HIPAA and GDPR guidelines regarding protected health information (PHI).
3. **Mitigation of Algorithmic Bias**: Balanced class distributions and CLAHE contrast normalization eliminate demographic and scanner-specific signal biases.

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11. CONCLUSION & FUTURE SCOPE

NeuroVision AI provides a complete, modern, and clinically grounded artificial intelligence solution for brain tumor detection. By combining state-of-the-art transfer learning with EfficientNetB0, explainable AI through Grad-CAM, and automated medical reporting with ReportLab, the platform delivers exceptional accuracy while solving the fundamental challenge of clinical interpretability.

Future Roadmap

1. **3D Multi-Planar Volumetric Reconstruction**: Extending from 2D axial slices to 3D T1-CE, T2-FLAIR volumetric segmentation using 3D U-Net architectures.
2. **DICOM PACS Integration**: Implementing direct DICOM network protocol adapters (C-STORE, C-FIND) to interface natively with hospital Picture Archiving and Communication Systems (PACS).
3. **Federated Learning Support**: Enabling multi-hospital decentralized model training to continually learn from new patient cases while preserving data residency within healthcare firewalls.

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